

# Andi Zhang

PH.D. CANDIDATE IN COMPUTER SCIENCE

Trinity College, Cambridge, CB2 1TQ, United Kingdom

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## Education

### University of Cambridge

PH.D. IN COMPUTER SCIENCE

- Supervised by Dr. Damon Wischik.

Cambridge, United Kingdom

Oct. 2019 - Jan. 2024 (expected)

MAST. IN MATHEMATICAL STATISTICS

Oct. 2018 - Jun. 2019

- Mathematical Tripos Part iii. Relevant modules:  
Statistical Learning in Practise, Convex Optimization, Astrostatistics  
Bayesian Modelling and Computation, Modern Statistical Methods

M.PHIL. IN ADVANCED COMPUTER SCIENCE

Oct. 2017 - Jun. 2018

- Relevant modules:  
Machine Learning for Natural Language Processing, Algebraic Path  
Deep Learning for Natural Language Processing, Advanced Functional Programming  
Probabilistic Machine Learning

### University of Manchester

B.SC. (HONS) IN COMPUTER SCIENCE AND MATHEMATICS

Manchester, United Kingdom

Sept. 2014 - Jun. 2017

- Relevant modules:  
Machine Learning, Quantum Computing, Compilers  
Convex Optimization, Probability, Statistical Methods  
Functional Analysis, Topology, Number Theory

## Experience

### Kirin AI group, Huawei

RESEARCH INTERN (PART-TIME) | MENTOR: MR. ARNAU RAVENTOS

- Working on video frame interpolation and AI global illumination.

Cambridge, United Kingdom

Nov. 2021 - Oct. 2022

### Noah's Ark Lab, Huawei

RESEARCH INTERN | MENTOR: DR. YITONG SUN

- Worked on deep learning uncertainty, 3 papers are published during the internship.

Shenzhen / Beijing, China

Sept. 2020 - Sept. 2021

### Trinity College, University of Cambridge

COURSE SUPERVISOR (TEACHING ASSISTANT)

- Supervised IA Algorithms (2020 Lent), IA Machine Learning (2020 Lent)  
and IA Introduction to Probability (2020 Easter)

Cambridge, United Kingdom

Oct. 2019 - Jun. 2021

### MRC Biostatistics Unit, University of Cambridge

RESEARCH ASSISTANT | SUPERVISOR: DR. SOFIA VILLAR AND DR. DAVID ROBERTSON

- Built a simulation system for multi-armed bandit problem (MABP) in clinical trial.
- Presented a new test statistic for binary response MABP.

Cambridge, United Kingdom

Jul. 2019 - Sept. 2019

### Tencent (WeChat AI)

R & D INTERN | MENTOR: DR. JIE ZHOU

- Concentrated on reinforcement learning and evolution strategy.

Beijing, China

Jul. 2018 - Sept. 2018

### Amazon (Alexa)

APPLIED SCIENTIST INTERN | MENTOR: DR. EMILIO MONTI

- Concentrated on Neural Machine Translation (Sequence to sequence model).

Cambridge, United Kingdom

Jun. 2017 - Sept. 2017

- Assisted in implementing general stance controller for the 3D printable robotic creatures.
- Helped reproduce the prototype animation system.

## Publications

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### Out-of-Distribution Detection with Class Ratio Estimation

Mingtian Zhang\*, Andi Zhang\*, Tim Z. Xiao, Yitong Sun, Steven McDonagh

<https://arxiv.org/abs/2206.03955>

### On the Out-of-distribution Generalization of Probabilistic Image Modelling

Mingtian Zhang\*, Andi Zhang\*, Steven McDonagh

In Neural Information Processing Systems annual meeting (NeurIPS), 2021

### Measuring Uncertainty in Signal Fingerprinting with Gaussian Processes Going Deep

Ran Guan, Andi Zhang, Mengchao Li, Yongliang Wang

In International Conference on Indoor Positioning and Indoor Navigation (IPIN), 2021

### Towards dynamic and scalable active learning with neural architecture adaption for object detection

Fuhui Tang, Dafeng Wei, Chenhan Jiang, Hang Xu, Andi Zhang, Wei Zhang, Hongtao Lu, Chunjing Xu

In The British Machine Vision Conference (BMVC), 2021

### Solipsistic Reinforcement Learning

Mingtian Zhang\*, Peter Noel Hayes\*, Tim Z. Xiao, Andi Zhang, David Barber

In Self-Supervision for Reinforcement Learning Workshop - International Conference on Learning Representations (ICLR), 2021

## Honors & Awards

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### The Williams/Kilburn Medal for outstanding final year student

Manchester, United Kingdom

School of Computer Science, University of Manchester

Jul. 2017

The medal is awarded on an occasional basis to students of exceptional distinction whose performance throughout their undergraduate course has been of outstanding merit.

### University awards for outstanding academic achievement

University of Manchester

The University will consider the top 0.5% of undergraduates (nominated by each School) for this award.

### Professors' Prize for the best third year student (joint honor)

School of Computer Science, University of Manchester

Prize for the Top 1 student.

### Professors' Prize for the best second year student

School of Computer Science, University of Manchester

Sept. 2016

Prize for the Top 1 student.

### Michael Jealous memorial prize for best first year student

School of Computer Science, University of Manchester

Sept. 2015

Prize for the Top 1 student.